

Nature's Way Soil - Comprehensive Website Improvements

Overview

This document outlines all the improvements made to natureswaysoil.com to enhance user experience, increase conversions, and provide educational content about soil health.

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Hero Video Implementation

Overview

Created an educational hero video explaining soil symbiotic relationships and how synthetic fertilizers disrupt natural processes.

Features

- **Scientific Accuracy:** Based on peer-reviewed research about mycorrhizal fungi and soil microbomes
- **Getty Images:** Professional stock footage through Pictory API
- **Engaging Narrative:** 90-second educational content that converts viewers
- **Key Statistics:**
 - 90% of plants rely on mycorrhizal fungi
 - 84% reduction in beneficial microbes from synthetic fertilizers
 - Natural nutrient exchange mechanisms explained

Component Location

- `components/HeroVideoSection.tsx`
- Integrated into `pages/index.tsx`

Video Script

Located at: `scripts/hero_video_script.txt`

Product Videos

Overview

Individual educational videos for each product explaining:

- The problem the product solves
- Scientific basis for how it works
- Key benefits and features
- Application instructions

Implementation

- **Video Generation:** Automated using Pictory API
- **Script Generation:** OpenAI GPT-4 creates product-specific scripts
- **Component:** `components/ProductVideoPlayer.tsx`
- **Integration:** Added to all product pages

Video Mapping

Videos are mapped in `config/video_urls.json` with format:

```
{
  "hero": "https://pictory.ai/video/...",
  "B0D9HT7ND8": "https://pictory.ai/video/...",
  "B0DXP97C6F": "https://pictory.ai/video/..."
}
```

Enhanced Chat Widget

Overview

AI-powered chat widget using OpenAI GPT-4 for intelligent customer support.

Features

- **Real-time Responses:** Instant answers to customer questions
- **Product Knowledge:** Trained on all product information
- **Soil Science Expertise:** Can explain complex concepts simply
- **Conversation History:** Maintains context across messages
- **Session Tracking:** Logs all conversations to Supabase

Components

- `components/EnhancedChatWidget.tsx` - Frontend UI
- `pages/api/chat.ts` - Backend API endpoint
- `lib/supabase_client.ts` - Database logging

System Prompt

The AI assistant is trained to:

- Explain soil microbiomes and mycorrhizal fungi
- Compare organic vs synthetic fertilizers
- Recommend products based on customer needs

- Provide application instructions
- Troubleshoot plant and soil issues

Usage

```
// Automatically included on all pages
import EnhancedChatWidget from '@components/EnhancedChatWidget'

<EnhancedChatWidget />
```

Email Automation

Overview

Sophisticated email marketing system using Resend API for lead nurturing and customer engagement.

Email Types

1. Welcome Email

- Sent immediately upon signup
- Includes 10% discount code
- Links to products and educational content
- Introduces the brand story

2. Abandoned Cart Email

- Triggered when items left in cart
- Lists cart items with images
- Reminds about free shipping threshold
- Includes urgency messaging

3. Educational Drip Campaign

- Day 1: Understanding Your Soil Microbiome
- Day 3: The Problem with Synthetic Fertilizers
- Day 5: Making the Switch to Organic
- Builds trust and educates customers

4. Order Confirmation

- Sent after successful purchase
- Order details and tracking
- Thank you message
- Cross-sell opportunities

Components

- `lib/resend_client.ts` - Email sending functions
- `components/AdvancedEmailCapture.tsx` - Signup forms
- `pages/api/email-capture.ts` - API endpoint

Email Capture Features

- **Multiple Placement:** Homepage, product pages, exit intent

- **Incentives:** Discount codes, free shipping
 - **Urgency:** Countdown timers
 - **Trust Indicators:** Security badges, no spam promise
 - **A/B Testing Ready:** Different headlines and offers
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Supabase Integration

Overview

PostgreSQL database for storing customer data, analytics, and engagement metrics.

Database Schema

Tables

email_captures

- id (UUID, **primary key**)
- email (VARCHAR)
- **source** (VARCHAR) - **Where** they signed up
- metadata (JSONB) - Additional context
- created_at (**TIMESTAMP**)

chat_logs

- id (UUID, **primary key**)
- session_id (VARCHAR)
- message (TEXT)
- response (TEXT)
- user_email (VARCHAR, optional)
- created_at (**TIMESTAMP**)

product_views

- id (UUID, **primary key**)
- product_id (VARCHAR)
- session_id (VARCHAR)
- created_at (**TIMESTAMP**)

popular_products (Materialized View)

- product_id (VARCHAR)
- view_count (**INTEGER**)
- unique_viewers (**INTEGER**)

Functions

- `captureEmail()` - Store email signups
- `logChat()` - Record chat conversations
- `trackProductView()` - Track product page visits
- `getPopularProducts()` - Get trending products

Migration

Run the migration: `supabase/migrations/001_initial_schema.sql`

Conversion Optimization Features

1. Inventory Tracker

Component: `components/InventoryTracker.tsx`

Features:

- Real-time stock levels
- Urgency messaging for low stock
- Out of stock notifications
- Color-coded alerts (green/yellow/red)

2. Social Proof Banner

Component: `components/SocialProofBanner.tsx`

Features:

- Recent purchase notifications
- Customer names and locations
- Product purchased
- Time since purchase
- Auto-rotating every 15 seconds

3. Personalized Recommendations

Component: `components/PersonalizedRecommendations.tsx`

Features:

- Based on viewing history
- Popular products algorithm
- Related products by category
- Dynamic product cards

4. Advanced Email Capture

Component: `components/AdvancedEmailCapture.tsx`

Features:

- Multiple incentive options
- Countdown timers
- Trust indicators
- Success animations
- Error handling

5. Product Bundles

Component: `components/ProductBundles.tsx`

Features:

- Frequently bought together
- Bundle discounts

- One-click add to cart
 - Savings calculator
-

Setup Instructions

1. Environment Variables

Create `.env.local` with:

```
# Supabase
NEXT_PUBLIC_SUPABASE_URL=your_supabase_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_supabase_anon_key

# OpenAI
OPENAI_API_KEY=your_openai_api_key

# Resend
RESEND_API_KEY=your_resend_api_key

# Pictory (for video generation)
PICTORY_API_KEY=your_pictory_api_key
PICTORY_CLIENT_ID=your_pictory_client_id
PICTORY_CLIENT_SECRET=your_pictory_client_secret

# Site URL
NEXT_PUBLIC_SITE_URL=https://natureswaysoil.com
```

2. Supabase Setup

1. Create a Supabase project
2. Run the migration: `supabase/migrations/001_initial_schema.sql`
3. Enable Row Level Security policies
4. Copy the URL and anon key to `.env.local`

3. Resend Setup

1. Sign up at resend.com
2. Verify your domain
3. Create an API key
4. Add to `.env.local`

4. OpenAI Setup

1. Get API key from platform.openai.com
2. Add to `.env.local`
3. Monitor usage in OpenAI dashboard

5. Pictory Setup

1. Sign up at pictory.ai
2. Get API credentials
3. Add to `.env.local`

6. Install Dependencies

```
npm install  
# or  
yarn install
```

7. Run Development Server

```
npm run dev  
# or  
yarn dev
```

8. Build for Production

```
npm run build  
npm start
```

API References

Chat API

Endpoint: `/api/chat`

Method: POST

Request:

```
{  
  "message": "What's the best fertilizer for tomatoes?",  
  "sessionId": "session_123",  
  "history": [  
    {"role": "user", "content": "Hello"},  
    {"role": "assistant", "content": "Hi! How can I help?"}  
  ]  
}
```

Response:

```
{  
  "response": "For tomatoes, I recommend our Organic Tomato Fertilizer..."  
}
```

Email Capture API

Endpoint: `/api/email-capture`

Method: POST

Request:

```
{
  "email": "customer@example.com",
  "source": "homepage_bottom",
  "metadata": {
    "incentive": "10% off",
    "timestamp": "2025-10-05T12:00:00Z"
  }
}
```

Response:

```
{
  "success": true,
  "message": "Email captured successfully"
}
```

Helper Scripts

OpenAI Helper

Location: `scripts/openai_helper.py`

Generate content:

```
python scripts/openai_helper.py "Create a product description for..."
```

Pictory Helper

Location: `scripts/pictory_helper.py`

Create video:

```
python scripts/pictory_helper.py script.txt "Video Title"
```

Generate All Videos

Location: `scripts/generate_all_videos.py`

Generate all product videos:

```
python scripts/generate_all_videos.py
```

Performance Optimizations

Image Optimization

- WebP format for modern browsers
- Lazy loading for below-fold images
- Responsive images with srcset

- CDN delivery

Code Splitting

- Dynamic imports for heavy components
- Route-based code splitting
- Lazy load chat widget

Caching Strategy

- Static assets: 1 year
- API responses: 5 minutes
- Product data: 1 hour

Analytics & Tracking

Events Tracked

1. Email captures (by source)
2. Chat interactions
3. Product views
4. Add to cart
5. Purchase completions
6. Video plays

Supabase Queries

```
-- Most popular products (last 30 days)
SELECT * FROM popular_products LIMIT 10;

-- Email capture rate by source
SELECT source, COUNT(*) as captures
FROM email_captures
WHERE created_at > NOW() - INTERVAL '7 days'
GROUP BY source;

-- Chat engagement
SELECT COUNT(DISTINCT session_id) as unique_sessions
FROM chat_logs
WHERE created_at > NOW() - INTERVAL '7 days';
```

Future Enhancements

Planned Features

1. **A/B Testing Framework:** Test different headlines, CTAs, layouts
2. **SMS Notifications:** Order updates and promotions
3. **Loyalty Program:** Points for purchases and referrals
4. **Quiz Builder:** “Find Your Perfect Fertilizer” quiz
5. **Video Testimonials:** Customer success stories

6. **Live Chat:** Real-time support during business hours
7. **Subscription Service:** Auto-delivery of products
8. **Mobile App:** iOS and Android apps

Technical Improvements

1. **Server-Side Rendering:** Improve SEO and performance
 2. **Edge Functions:** Faster API responses
 3. **GraphQL API:** More efficient data fetching
 4. **Progressive Web App:** Offline functionality
 5. **Advanced Analytics:** Heatmaps, session recordings
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Troubleshooting

Common Issues

Chat not responding

- Check OpenAI API key is valid
- Verify API endpoint is accessible
- Check browser console for errors

Emails not sending

- Verify Resend API key
- Check domain verification
- Review Resend dashboard for errors

Videos not loading

- Check video URLs in config/video_urls.json
- Verify Pictory API credentials
- Ensure videos are publicly accessible

Database errors

- Verify Supabase credentials
 - Check RLS policies are correct
 - Review Supabase logs
-

Support & Maintenance

Regular Tasks

- Monitor API usage and costs
- Review chat logs for quality
- Update product videos quarterly
- Refresh email templates seasonally
- Analyze conversion metrics monthly

Contact

For technical support or questions about this implementation:

- Email: dev@natureswaysoil.com

- Documentation: /docs/
 - GitHub Issues: github.com/natureswaysoil/coplit-built/issues
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Credits

Technologies Used

- **Next.js**: React framework
- **TypeScript**: Type safety
- **Supabase**: Database and auth
- **OpenAI**: AI chat responses
- **Resend**: Email delivery
- **Pictory**: Video generation
- **Tailwind CSS**: Styling
- **Vercel**: Hosting

Research Sources

- Nature Communications: Soil microbiome research
 - Frontiers in Plant Science: Mycorrhizal studies
 - Various peer-reviewed journals on soil health
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